

Technology Innovation HUB Program Update

For the **National Football League**

Program Update | January 8, 2024

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Meeting Agenda

1

Background

Recap of Previous Presentation
Explanation of Value Zone

2

Innovation HUB

Program Design and Criteria
Governance Structure

3

Operating Model of the Innovation Hub

Execution Phase Gates

4

Pilot Project Updates

Optical Tracking Progress
NFL Player View Project
Developments
NLP / Multi-modal AI Milestones

5

NLP / Multi-modal AI Pitch

Introduction
Business Value



4 Zones Model: Business Value Creation



- **Disruptive Innovations:** New business models and operating models
- **Sustaining Innovations:** Extension and improvements to already existing innovations inside the organizations



Program Design & Criteria

Program Design

- VC Model of Investment
- Full visibility on objectives and progress of each project
- Centralized decision making on which projects to fund

Established Success Criteria

- Funding decision based on rigorous screening and evaluation process
- Funding linked to stage gates or milestones
- Control of funding based on milestone-based success criteria
- Multiple exit options for sponsors based on process success



Governance Structure

The Governance Committee's mission is to steer the Innovation Hub program through collaborative design, guiding league-wide decisions for maximum impact.

Executive Steering Committee

EVPs

Program Execution Committee

SVPs - Information Technology, Player Health and Safety, Football Operations,
NFL Media, Data & Analytics, Finance, and Sponsorship

Innovation Tech Advisory Committee

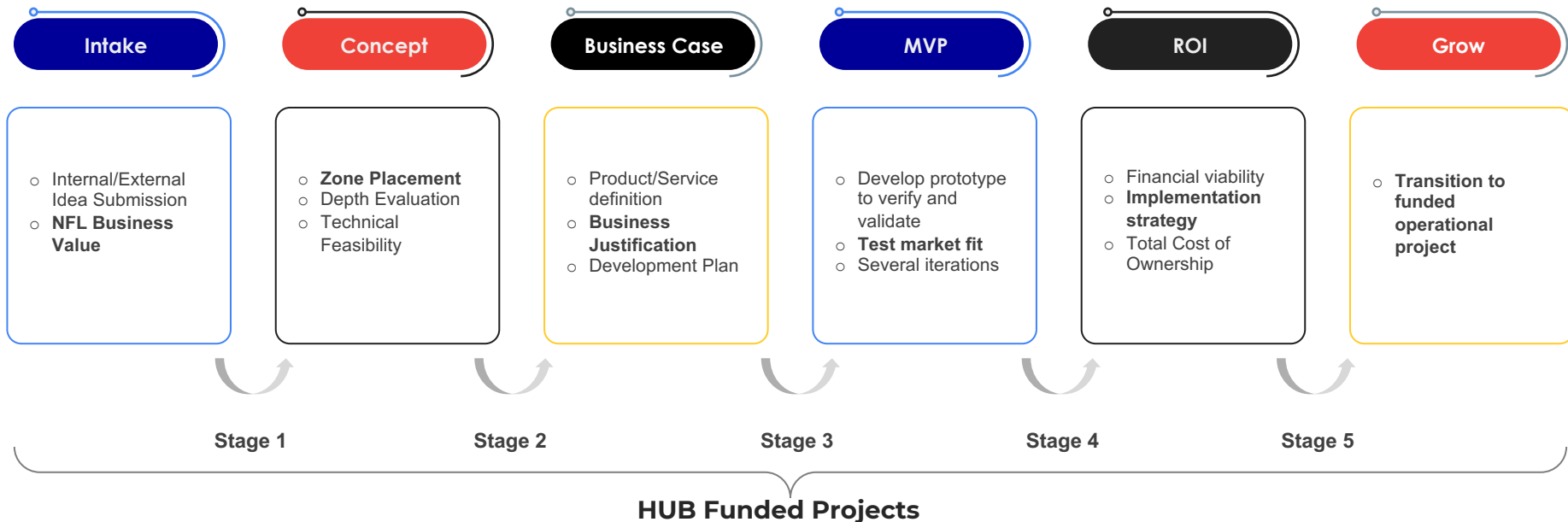
NFL Technical Working Group

Innovation HUB PMO

Intake, Prioritization, & Program Management



HUB Execution Gates or Operation Model





POC Projects

1

Optical Tracking

2

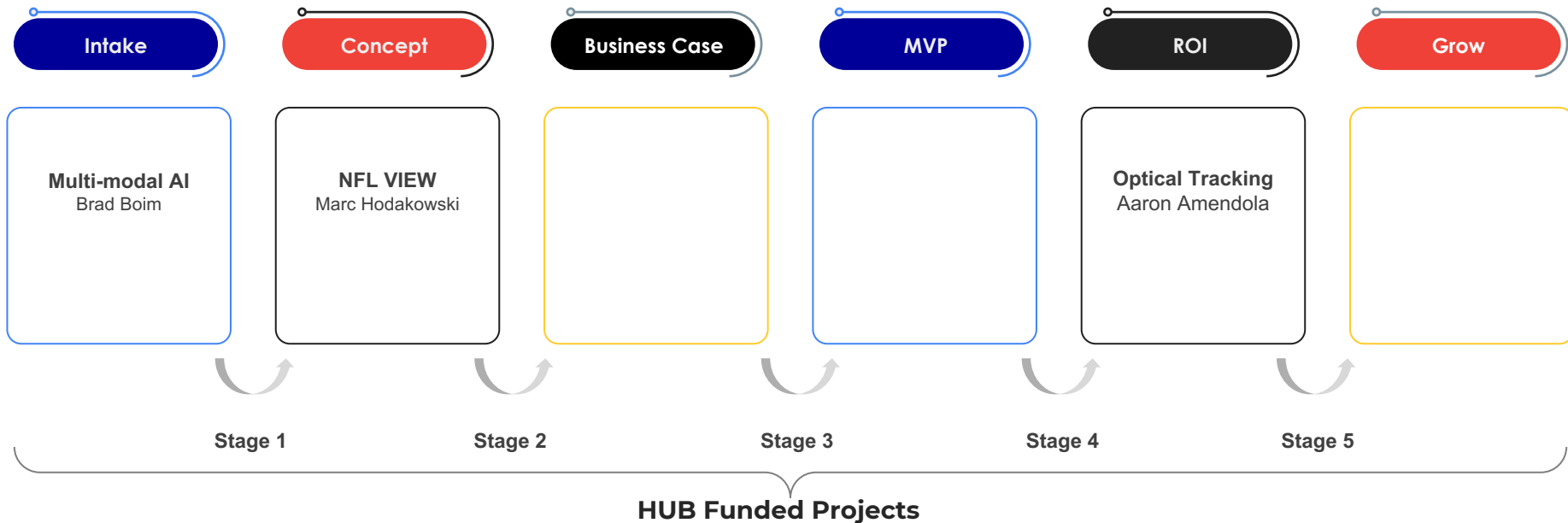
NFL Player View

3

Multi-Modal AI



Project Stage Placements



Optical Tracking

ROI



Overall Scope: The optical tracking scope aims to identify an innovative solution for line-to-gain detection and other officiating use cases. Optical tracking will be able to inform real-time officiating.

Current State:

- **Conducting 3 POCs to determine the accuracy of each demo.**
Vendors: HawkEye, ChyronHego, Genius/Second Spectrum.
- **Stadium Evaluations:** Each vendor is undergoing evaluations and testing in stadiums upon approval
- **Vendor Scorecard & Partner Screening:** A vendor scorecard has been developed to assess compatibility and are actively screening potential partners during the Proof of Concept (POC) phase.



Optical Tracking, Cont.

Use Case

- **Line-to-Gain:** Utilize optical tracking technology to improve the accuracy of line-to-gain detection.
- **Game Presentation:** Optimized data utilization for all games (Including in-stadium and broadcast executions).

Next Steps

- Data accuracy comparison among vendors, aiming for consistently achieving $\frac{1}{2}$ inch accuracy or better ($\frac{1}{2}$ or less).
- Identify the vendor that best aligns with the goals identified through **RFI Vendor Evaluation Scorecard**.



Optical Tracking

Vendor Assessment

	HawkEye <i>Current Vendor</i>	Genius/Second Spectrum <i>Current Vendor</i>	ChyronHego <i>New Vendor</i>
Overview	Technology is currently deployed in 3 stadiums (LV, MIA, NY)	SoFi Stadium (LA)	Technology has been deployed at 5 stadiums for MLS soccer (ATL, CAR, CHI, NE, SEA)
Update	Data from testing has been analyzed	Installation has been completed in SoFi Stadium (LA)	NDA has been signed
Next Steps	A review with the Competition Committee and the Future of Football Committee scheduled during Combine (Feb 27-Mar 1)	Tech evaluation scheduled for the January 7th game	Validating solution internally before enabling system for January 7th

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NFL View

Concept



Overall Scope: NFL View's primary emphasis is player health and safety. Their concentration encompasses player-related elements such as player speed, impact force and injury prevention.

Current State

- Enhance data collection for player health and safety.
- Three years of testing and utilizing original Intel cameras for player movement tracking.



NFL View

Concept

Use Case

- **Player Health and Safety:** Proactively preventing injuries by identifying game events that enhance player safety.
- **Identifying Game events:** Identifying game events can help improve player engagement and enhance the overall gaming experience.
- **Skeletal tracking:** This is implemented to proactively enhance player safety and performance.

Next Steps

- Collaborating with the PHS team to create a comprehensive plan for business testing, cost analysis, and project timeline development.



NFL View



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Strategic Alignment

- New Technology
- Non-Game Media
- International Expansion

POC Completed

June - October

100 hours of NFL content indexed, 10 taxonomies fine-tuned]

Phase One Goals – April 2024

- Direct integration of TwelveLabs API with Vidispine Media Asset Management
- Build, test and refine integration and validate results
- Identify and engage power users to test and provide feedback

Innovation Hub - Funding Request

Designed for:

MutiModal AI

Designed by:

Meloni Boatswain

Date:

12/12/2023

Version:

1.1

Problem  - Current media search process is inefficient and time consuming - Most content is not logged, buried deep in the archive, and impossible to find. - Human logging process is antiquated, costly and error prone (\$25K / year)	Solution  <i>Develop an automated solution leveraging Multi-Modal AI via the TwelveLabs API to allow Natural Language search across our content library at a scale.</i>	Unique Value Prop  - Develop a natural language (AI/ML) search tool for media assets within the NFL to unlocking previously unavailable content - Train model for a customized taxonomy for exclusive use by the NFL	Expected Outcome  - Faster search efficiency - Improved discovery of our media archive, uncovering assets previously buried in the archive - Quick adoption, as it utilizes UI tools already in use	Customer Segments  - Internal Content Creators - NFL Media Producers - Compliance
Existing Alternatives  - Manual asset tagging - Not logging content at all - Traditional AI methods	Key Metrics  - Accuracy of search results - Search time reduced from minutes/hours to seconds - Is the indexing pipeline sustainable? - Faster user adoption	High-Level Concept  - TwelveLabs multi-modal AI extracts multiple features from video such as time, objects, text in video, conversation, faces, and actions, synthesizing this information into vectors. - Vectors enable fast, semantic and scalable search using natural language.	Next Steps  - Direct integration into the NFL's existing MAM system - Develop indexing and search pipeline - Fine-Tune NFL Taxonomies - Deployment and testing during 2023 Playoffs to determine capabilities and benefits	Early Adopters  - NFL Media Teams (L.A., NJ, NY) - NFL Game Operations
Cost Structure  Integration, development, testing, and initial deployment: - ARVATO SYSTEMS NORTH AMERICA \$15,000 - TwelveLabs \$36,300 Total: \$51,300			Value Proposition  - Cost savings from increased efficiency and accuracy. - Exploration of Gen AI technology targeting potential applications within the NFL. - Gaining leverage from existing investments in our media asset library utilizing emerging Gen AI technology.	

Multimodal AI Pitch



Pitch Feedback



Next Steps



Disruptive Innovation Zone Placements

Incubation Zone

- Develop an MVP:
 - Focus on developing a minimum viable product (MVP) quickly and efficiently to gain early feedback and validate the core concept.
- Milestones:
 - MVP Prototype completed within X weeks/months.
 - User testing conducted with Y participants.
 - Key learnings and insights documented.
- Success Criteria:
 - MVP demonstrates core functionality and potential value.
 - User feedback validates the problem and need for the solution.
 - Clear path identified for further development based on testing results.

Transformation Zone

- Conduct a pilot project:
 - Implement the idea on a smaller scale to test its feasibility and impact within a controlled environment.
- Milestones:
 - Pilot project plan established within X weeks/months.
 - Pilot launched and executed successfully for Y timeframe.
 - Data and metrics collected to measure impact on KPIs and other critical areas.
- Success Criteria:
 - Pilot proves the idea's scalability and ability to generate desired outcomes.
 - Financial modeling demonstrates strong ROI potential.
 - Clear plan for full-scale launch developed based on pilot learnings.

Sustaining Innovation Zone Placements

Performance Zone

- Integrate and deploy:
 - Seamlessly integrate the idea into existing NFL operations and performance metrics.
- Milestones:
 - Implementation plan developed within X weeks/months.
 - Idea integration completed with minimal disruption to existing systems.
 - Training and support provided to relevant stakeholders.
- Success Criteria:
 - Measurable improvement in target KPIs (e.g., increased fan engagement, revenue growth, contribution margins).
 - Positive user feedback and adoption from internal stakeholders.
 - Successful integration with existing systems and infrastructure.

Productivity Zone

- Implement and optimize:
 - Implement the idea to automate tasks, streamline processes, and optimize resource allocation.
- Milestones:
 - Implementation plan developed within X weeks/months.
 - Key processes identified for optimization.
 - Technology infrastructure established to support automation.
- Success Criteria:
 - Demonstrated cost savings or resource optimization based on implemented changes.
 - Improved efficiency and reduced waste in target processes.
 - Positive feedback from stakeholders impacted by the changes.

STRATEGIC PLANNING FOCUS



*League goals in no particular order